Maperitive

Making Your Own Maps

by Igor Brejc

Contents

- Goals
- Main features and capabilities
- Challenges
- Future plans
- Experimental stuff

Maperitive & Me

- Living in Maribor, Slovenia
- Professional software developer
- Long-running love for cartography
- Joined OSM in 2007
- OSM speciality: hiking

Why Maperitive?

- Started as Srtm2Osm, Kosmos
- Existing solutions (Mapnik, Osmarender) lacking simplicity of use, immediacy and interactivity
- Wanted to have my own project to play with
- Kosmos needed a total overhaul (including a more googleable name)

Goals For Maperitive

- Simple to use...
- … and yet powerful enough...
- … to produce professional-looking maps
- Single-stop shop for making OSM maps
- Things should work out of the box
    - Convention over configuration

Main Features

- Rendering rules (special DSL)
- Embedded command-line interface
  - Scripting
- Web maps integration (browser-like)
- Integration with SRTM + custom DEMs
  - Contours, hillshading, hypsometric coloring
- Web map tiles generator
- Exporting to bitmap and SVG

Rendering Rules

- Specify features (what to render?)

```
boundary country : boundary=administrative
   AND (admin_level=2 OR admin_level=4)
   AND NOT natural=coastline
```

- Specify rules (how to render?)

```
target : aeroway line*
   define
      min-zoom : 9
      line-color : #9D9595
   if : aeroway line runway
      define
         line-width : 9:1;10:1;11:2;13:6;15:20
   else
      define
         line-width : 9:1;11:1;13:3;15:10
   draw : line
```

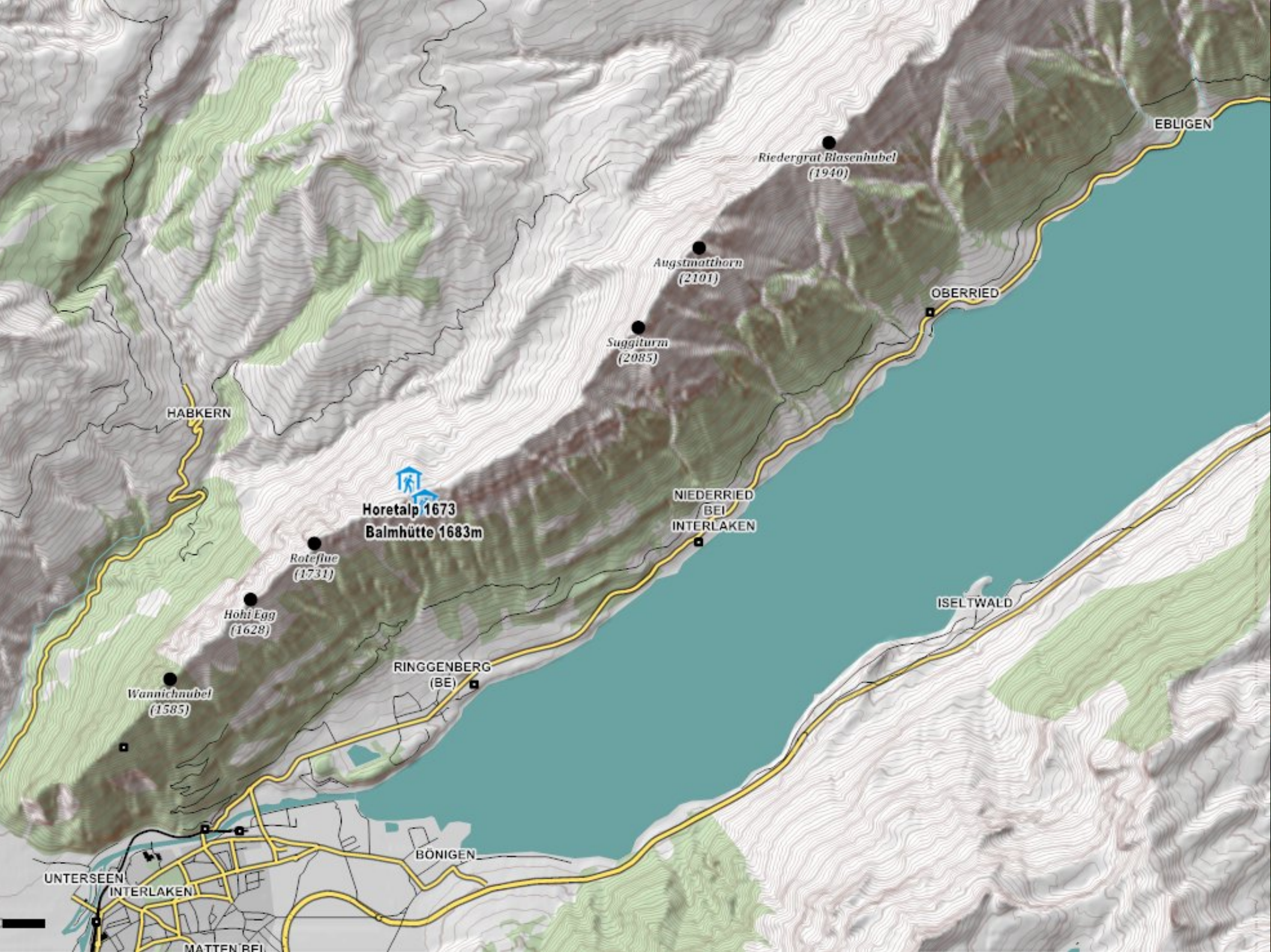
Commands And Scripts

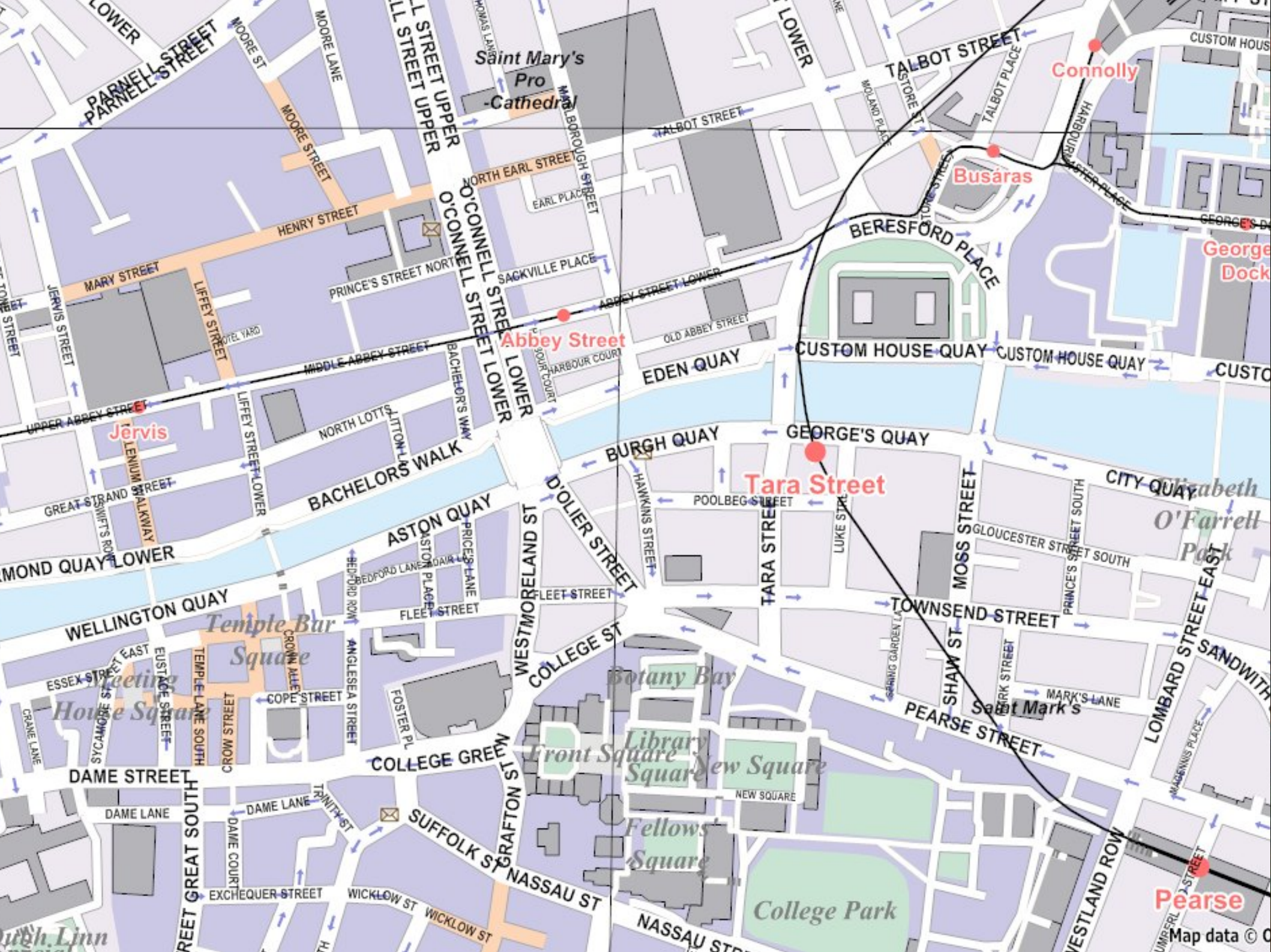
- Everything is available through the command line
- Common functions also through the GUI
- 81 commands currently available

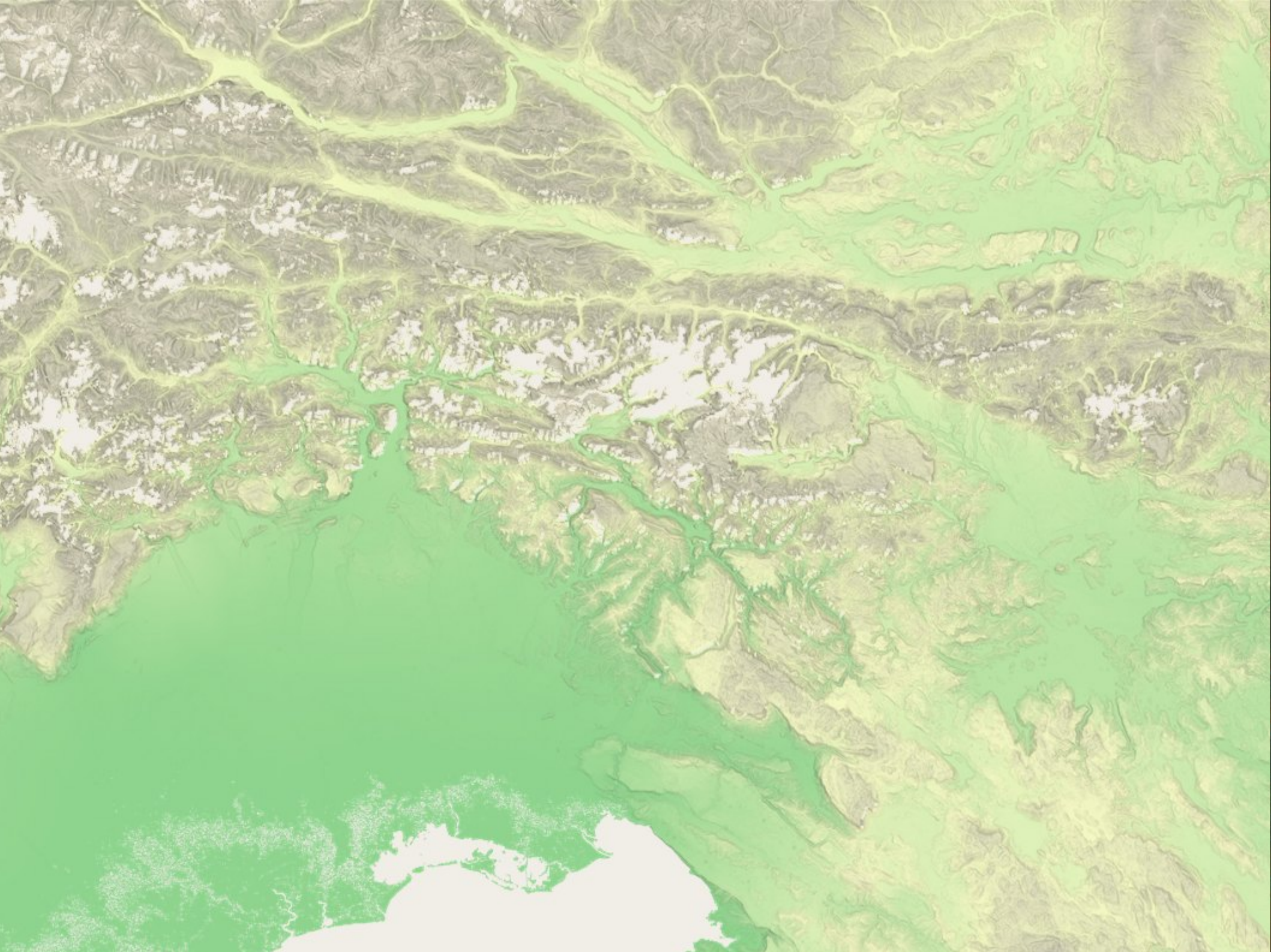
```
clear-map
set-setting name=map.decoration.grid value=False
use-script-dir
use-ruleset location=dublin.mrules
load-source Dublin.osm.bz2

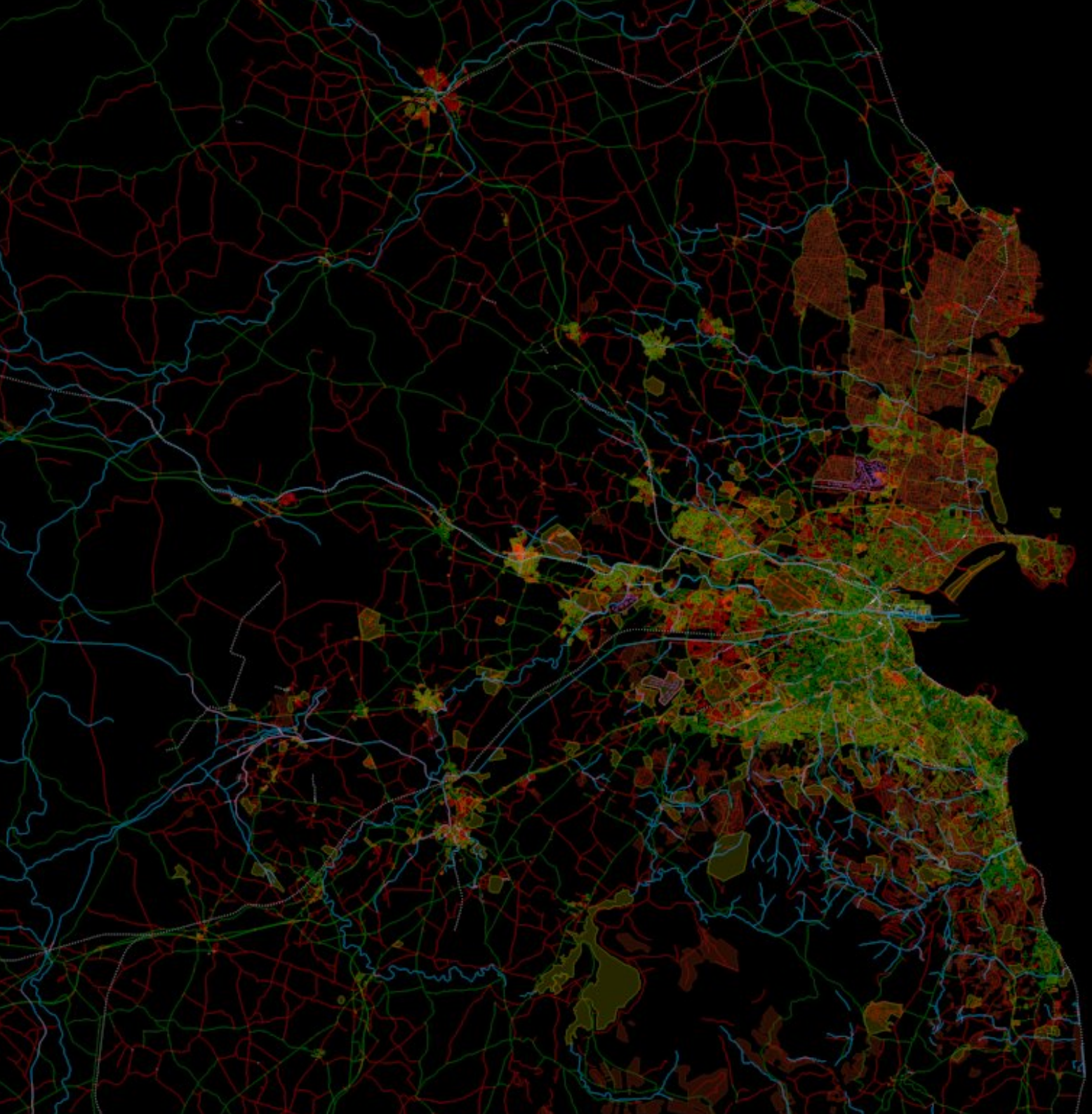
bounds-set -6.658312516764 64,53.3270920303852,
   -6.3469551584,53.5082149

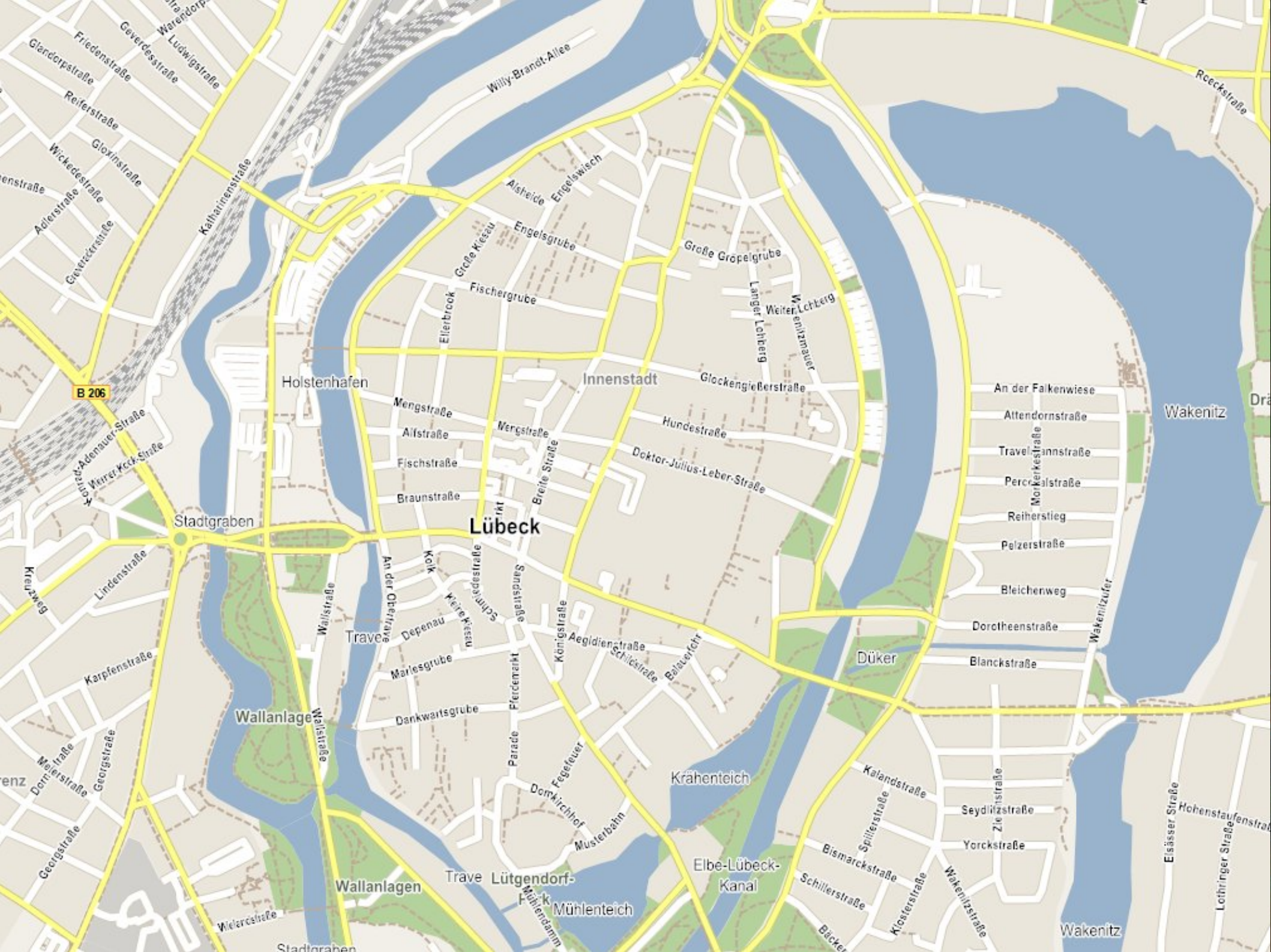
export-svg file=extra\Dublin-NW-illustrator.svg
   zoom=16 compatibility=illustrator
   ai-autorescale=true
```


EBLIGEN
Riedergrat Blasenhubel
(1940)
Augstmatthorn
(2101)
OBERRIED
Suggiturm
(2085)
HABKERN
Horetalp 1673
Balmhütte 1683m
NIEDERRIED
BEI
INTERLAKEN
Roteflue
(1731)
Höhi Egg
(1628)
ISELTWALD
RINGGENBERG
(BE)
Wannichnubel
(1585)
BÖNIGEN
UNTERSEEN
INTERLAKEN
MATTEN BEI


Saint Mary's Pro -Cathedral
Connolly
Busáras
Abbey Street
Jervis
Tara Street
Pearse
Temple Bar Square
Meeting House Square
Botany Bay
Library Square
Front Square
New Square
Fellows Square
College Park
Elizabeth O'Farrell Park
George Dock
PARNELL STREET
MOORE STREET
MOORE LANE
O'CONNELL STREET UPPER
O'CONNELL STREET LOWER
MARLBOROUGH STREET
TALBOT STREET
NORTH EARL STREET
HENRY STREET
MARY STREET
PRINCE'S STREET NORTH
SACKVILLE PLACE
ABBEY STREET LOWER
OLD ABBEY STREET
BERESFORD PLACE
CUSTOM HOUSE QUAY
EDEN QUAY
MIDDLE ABBEY STREET
UPPER ABBEY STREET
JERVIS STREET
LIFFEY STREET
LIFFEY STREET LOWER
NORTH LOTTS
BACHELORS WALK
BURGH QUAY
GEORGE'S QUAY
CITY QUAY
POOLBEG STREET
GREAT STRAND STREET
ASTON QUAY
D'OLIER STREET
HAWKINS STREET
TARA STREET
LUKE STREET
MOSS STREET
GLOUCESTER STREET SOUTH
PRINCE'S STREET SOUTH
ORMOND QUAY LOWER
WELLINGTON QUAY
FLEET STREET
WESTMORELAND ST
COLLEGE ST
TOWNSEND STREET
SHAW ST
PEARSE STREET
LOMBARD STREET EAST
SANDWITH
ESSEX STREET EAST
EUSTACE STREET
TEMPLE LANE SOUTH
COPE STREET
CROWN ALLEY
ANGLESEA STREET
FOSTER PL
MARK'S LANE
DAME STREET
COLLEGE GREEN
DAME LANE
DAME COURT
TRINITY ST
GRAFTON ST
SUFFOLK ST
NASSAU ST
EXCHEQUER STREET
WICKLOW ST
NEW SQUARE
Map data ©






Lübeck
Innenstadt
Holstenhafen
Stadtgraben
Wallanlagen
Krähenteich
Mühlenteich
Elbe-Lübeck-Kanal
Wakenitz
Düker
Trave
Willy-Brandt-Allee
Roeckstraße
Katharinenstraße
Adenauer-Straße
Weiter Kreck-Straße
B 206
Engelswisch
Engelsgrube
Große Gropelgrube
Große Kiesau
Fischergrube
Elierbrook
Langer Lohberg
Weiter Lohberg
Wallmauer
Glockengießerstraße
Hundestraße
Doktor-Julius-Leber-Straße
Mengstraße
Alfstraße
Fischstraße
Braunstraße
Breite Straße
Markt
Koberg
Schmiedestraße
Sandstraße
Königstraße
Aegidienstraße
Schildstraße
Balauerfohr
An der Obertrave
Depenau
Marlesgrube
Dankwartsgrube
Fischermarkt
Parade
Fegefeuer
Domkirchhof
Musterbahn
Wallstraße
Weiterstraße
Lindenstraße
Karpfenstraße
Meierstraße
Domstraße
Georgstraße
Mühlendamm
Lütgendorf-Fork
An der Falkenwiese
Attendornstraße
Travemannstraße
Percevalstraße
Reiherstieg
Pelzerstraße
Bleichenweg
Dorotheenstraße
Blanckstraße
Wakenitzufer
Morkerkestraße
Kalandstraße
Seydlitzstraße
Yorckstraße
Bismarckstraße
Schillerstraße
Spillerstraße
Klosterstraße
Wakenitzstraße
Elsässer Straße
Hohenstaufenstraße
Lothringer Straße

Challenges & Future Plans

- Interactivity vs quality vs quantity
  - Render quickly
  - Render with quality
  - Render big maps with lot of data
- Database support (PostGIS, spatialite)
- Using Direct2D for rendering

Challenges & Future Plans

- How to extract info from data?
- Poor support for OSM relations
- OSM data is raw data – it needs preprocessing before rendering
- Example: bus route relations
    - Multiple ways in a relation
    - Multiple relations in a way
    - Rendering a bus map without loss of information
- Example: OSM Mapnik map & transparent boundaries
    - Inconsistent rendering because several admin levels are rendered separately one atop the other

Challenges & Future Plans

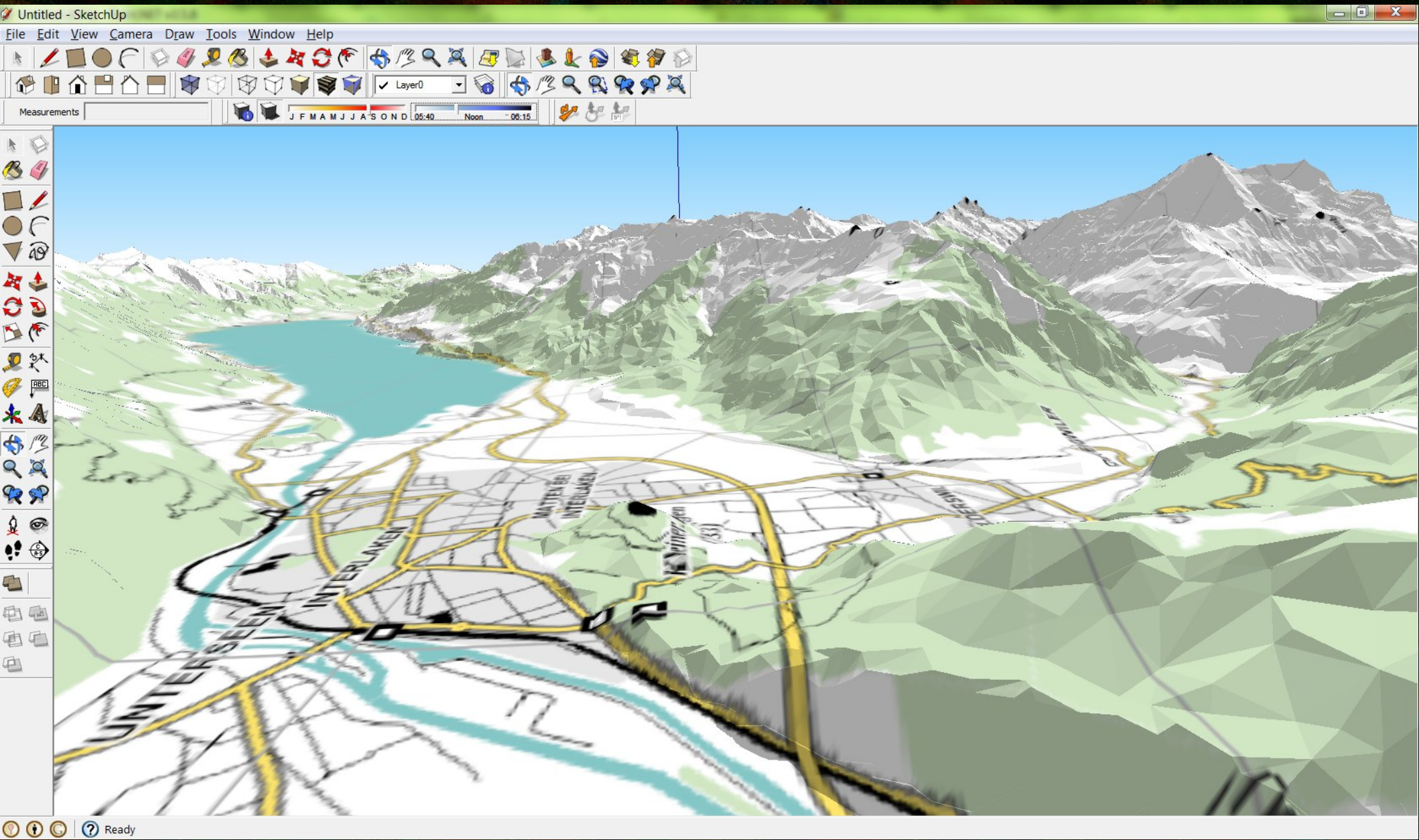
- Possible solutions:
- Transforming raw OSM data into more advanced data structures (like networks, graphs)
- More advanced toolset for data analysis and synthesis
- Perhaps even a new file storage format for these transformations
    - Or storing it into DB
- Python for scripting
- Extensibility (plug-ins)

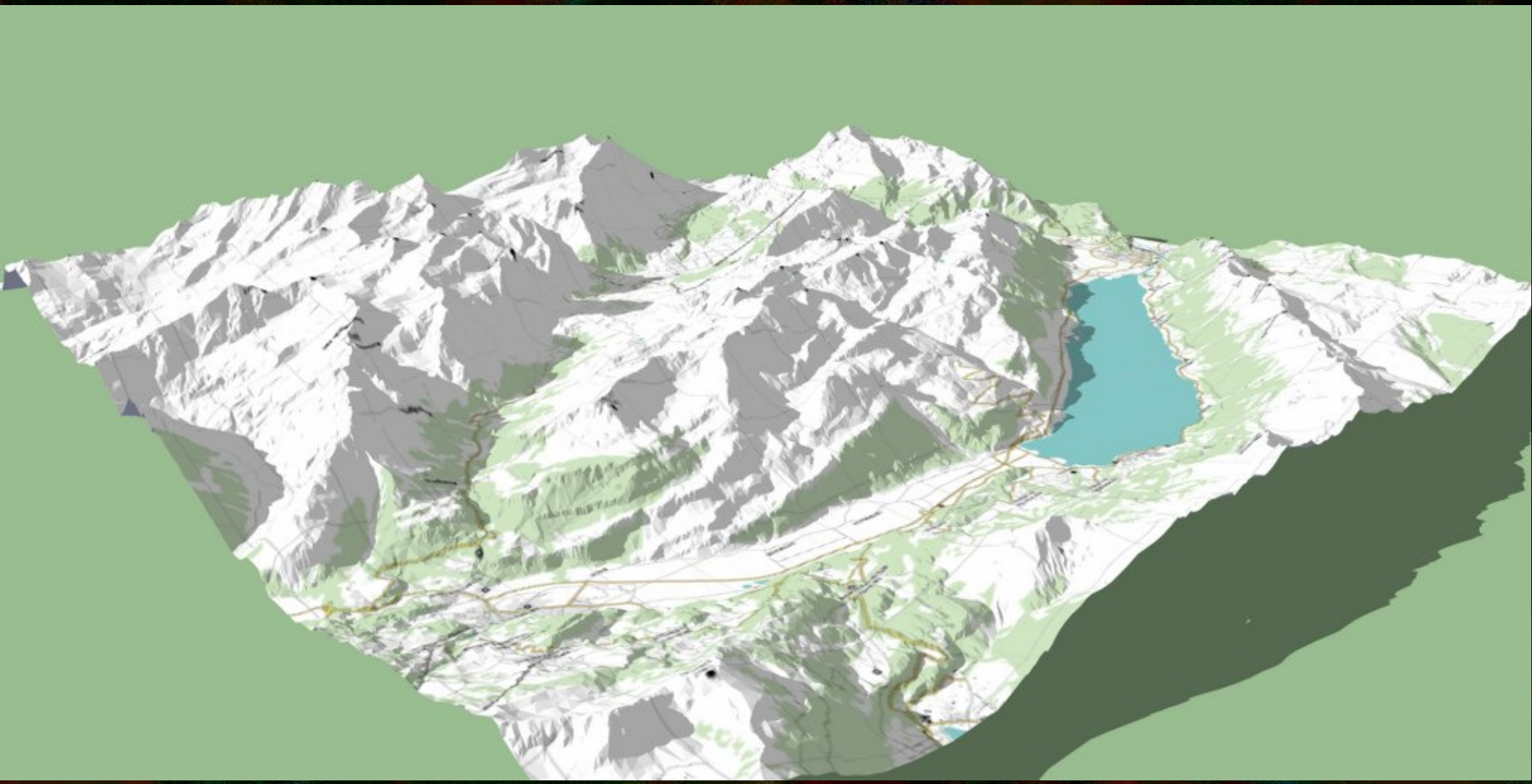
Challenges & Future Plans

- Map labeling improvements
    - Detecting overlaps, duplicates
    - Repeating labels for long streets
    - Abbreviations
- Support for shapefiles, GeoTIFFs
- Other projections, rotating the map
- Headless mode – no GUI
- Multi-core tiles generation
- Maperitive Remote – REST WS
- GUI improvements
- Map drawing editor?

Experimental

- 3D map export in COLLADA format
- Export to RaphaelJS


Untitled - SketchUp
File Edit View Camera Draw Tools Window Help
Layer0
Measurements
J F M A M J J A S O N D
05:40
Noon
06:15
UNTERSEEN
INTERLAKEN
MATTEN BEI INTERLAKEN
Ready



Discussion...